

Capstone Project: Personal Assistant

Project Overview

Introduction and Problem Statement

The capstone project serves as a culminating experience, allowing you to apply the knowledge and skills you have acquired throughout your studies to a real-world problem or challenge. This project aims to apply to a broader range of tasks in your day-to-day life, as you'll make a personal assistant augmented with agentic tools to help with the provided tasks. The application could range from booking airline tickets and hotel rooms for conference travel to simply checking and generating drafts for your emails; the choice of application is up to you, depending on the problem you want to solve.

Project Goals and Objectives

The primary goal of this capstone is to help demonstrate your capability to utilize the agentic tools and concepts obtained throughout the duration of the course.

Concepts	Description	Success Metric
Tool Calling	Utilize an agentic architecture to provide the large language model with tool calling abilities.	Agent is able to execute tool calls and utilize functions while understanding the output.
Reasoning Architectures and Chain of Thought	Apply a basic reasoning loop such as ReAct so that the agent is able to interact with its current environment.	The agentic system displays the capability to utilize reasoning and adequately recover from missteps.
Knowledge and Memory	Utilize a retrieval based memory mechanism, such as RAG or other proposed methods, to retain knowledge between tool calls.	The agentic system can recall past task executions or use a store of knowledge, such as API documentation or other data.

Concepts	Description	Success Metric
Further Reasoning	Further build up the reasoning loop using concepts such as Tree of Thought	The system can more clearly outline the agent's reasoning architecture and thought process.
Multi-agent Coordination	Develop the agentic system to leverage parallelism and specialized agents to assist with various tasks.	Demonstration of at least one agent to execute tasks.
Safety	Ensure that the agentic system is safe to deploy and that specified guardrails are in place to prevent unintended task execution.	Statement of potential unintended actions the system may execute and the steps taken to address this.

Deliverables

The final project deliverables will include:

- **Final Report:** A comprehensive document detailing the problem, methodology, results, and conclusions.
- **Presentation:** A formal presentation summarizing the project for the faculty and peers.